

Router Governance & the Ablation Evidence

Two artefacts, one vocabulary. The **ablation measures** (real models, **REAL** OpenRouter backend, 2 reps) where an LLM degenerates without or with bad epistemic state. The **governance layer enforces** (deterministic, synthetic fixtures, no model call) by turning those exact degeneration metrics into router gates. This document ties the two together — conservatively, with no claim beyond the data.

Sources: `ablation2_phase5_sonnet.json` (anthropic/claude-sonnet-4.5) & `ablation2_phase5_granite.json` (ibm-granite/granite-4.1-8b); Phase 5 = final long-document run.

1 · What the ablation found — recall & token cost

Condition	Recall (Sonnet 4.5)	Recall (Granite 4.1-8b)	mean input tokens
A — full context	0.88	0.76	18342
B — DESi state	1.00	0.96	372
B — auto-constructed	0.96	0.88	450
E — status-stripped (budget-matched)	1.00	0.94	379
R1 — BM25 retrieval	0.52	0.44	355
R2n — neural retrieval	0.08	0.10	355

DESi state is load-bearing — **B ≥ A at ~49× fewer tokens** (372 vs 18 342). **B ≈ E**: stripping the status typing under a matched token budget does not move recall, so the metadata typing is *not* the recall driver. Retrieval without state collapses (R1 lexical) or, neurally (R2n), all but vanishes.

2 · Where degeneration was measured (rate over reps)

Metric	B	E	R2n	A
coherence_without_continuity	0.00	0.00	0.80	0.00
confidence_while_wrong	0.00	0.00	0.60	0.00
loop_trap	0.00	0.00	0.40	0.00
contradiction_persistence	0.00	0.00	0.00	0.00
invalid_claim_reuse	0.00	0.00	0.00	0.00
bad_framing_nonrecovery	0.00	0.00	0.00	0.00

Rates shown for Sonnet 4.5. **R2n — neural retrieval without DESi state — is the toxic path**: fluent, but it loses the state and is confidently wrong (coherence-loss 0.80, confidence-while-wrong 0.60, loop-trap 0.40). On Granite, condition A also persists contradictions (0.60). B and E stay clean. This R2n column is exactly the case the router never answers blindly.

3 · The governance layer — DESi diagnoses, the router acts

A deterministic layer in `desi_router/governance/`. It consumes a read-only **DesiReport** (a projection of the Layer-9 EpistemicGapSnapshot), picks one of eight **epistemic modes** as a new axis *alongside* the existing tool/local/API routing, optionally builds a guarded preprompt, **verifies the answer after the fact**, and audits the decision. It never enforces inside DESi and never mutates persistent state — Layer-9's gate stays the authority; the router only decides whether an answer may *propose* an update and whether a verifier must pass.

Mode	When	Effect
state_slice	clean usable state, low/moderate risk	slice in prompt; verify if moderate
guarded	invalidated/superseded touched, open conflict, or high poisoning	guarded preprompt + verifier required; no update unless it passes
recovery	wrong frame already in the conversation + high poisoning	recover from the wrong frame using current valid state
retrieval	no usable DESi state, evidence-lookup task	BM25 / embedding / hybrid
anti_delphi	open conflict the answer would resolve, challenger exists	challenge before stabilising
ask_user	required user-specific state missing	request clarification
normal	low risk, no state needed	plain prompt; may propose update

4 · The bridge — same metric, now a gate

The governance tests carry **no recall number**; they assert pass/fail that the gate *fires*. The router verifier re-implements the ablation's degeneration metrics, so the link is metric-for-metric:

Ablation measures (empirical rate)	Router verifier check (enforces)	Governance test
coherence_without_continuity = 0.80 @R2n	coherence_without_continuity (warns)	test_coherence_without_continuity_warns...
confidence_while_wrong = 0.60 @R2n	stale_confident_answer (BLOCKS)	test_stale_confident_answer_with_no_state_blocks
invalid-claim reuse (wrong-slice phases)	invalid_claim_reuse (BLOCKS)	test_verifier_catches_invalid_claim_reuse
contradiction_persistence = 0.60 @A (Granite)	conflict_closure_without_evidence (BLOCKS)	test_open_conflict_closed_without_evidence
bad_framing_nonrecovery	→ routes to recovery_mode	test_high_poisoning_is_guarded_or_recovery

The routing also mirrors the recall table: the R2n situation (no state → collapse + degeneration) is precisely where `select_mode` refuses a blind answer — no usable state → **retrieval mode**; risky state → **guarded/recovery** + a required verifier.

5 · What this does and does not claim

- **Supported:** DESi state selection is load-bearing ($B \geq A$ at ~49× fewer tokens).
- **Supported:** wrong/absent state degenerates measurably (R2n), and the router routes around exactly that case.
- **Supported (enforcement):** 26 governance tests prove the verifier fires on the same failure modes the ablation measured — critical checks block, coherence only warns.
- **NOT claimed:** metadata governance is a recall effect — $B \approx E$ stands. This layer governs *behaviour around* the state, not extraction quality.
- **Limits:** the verifier is rule-based (token-overlap + cue words), sharing the ablation evaluator's paraphrase/negation blindness; risk thresholds are fixed, not calibrated.

Reproduce: `python -m desi_router.governance.demo · pytest tests/router_governance -q (26 tests) · ablation numbers from the two Phase-5 result JSONs above.`